DriveIQ

PostHog Analytics Implementation Spec

Technical handoff for instrumentation, event taxonomy, dashboards, privacy guardrails, and rollout QA.

Company: DriveIQ Technologies Ltd
Date: August 2026
Status: Ready for implementation

This document is built from a full codebase audit. It defines exactly what to track in PostHog, where to instrument it, and how to validate data quality before launch.

1. Objectives and success criteria

Goal: integrate PostHog so the team can make product and revenue decisions from trustworthy data, not guesswork.

- Measure full user journey: discovery -> engagement -> routing -> retention -> upgrade.
- Track free vs premium behavior with clear funnel breakpoints.
- Track system health (provider failures, timeouts, cache behavior) next to product metrics.
- Enforce privacy-safe event payloads and avoid accidental PII capture.

North-star reporting outputs

- Activation: first meaningful action rate within first session.
- Engagement: weekly active users and repeat usage by feature cluster.
- Conversion: paywall shown -> trial started -> subscription started.
- Retention: D1, D7, D30 by acquisition cohort and tier.
- Reliability: provider timeout/failure rates correlated with engagement drop.

Implementation rule

Every event must answer a business question. If an event does not drive a dashboard, alert, or decision, do not ship it.

2. Analytics architecture

Recommended design in this repo

1. Create src/services/analytics.ts wrapper with typed track(name, props), identify(userId), reset(), and page(screen).
2. Initialize PostHog in app/_layout.tsx only once; set persistence mode and app-level properties.
3. Call identify/reset from AuthProvider on auth state transitions.
4. Instrument user actions in app/index.tsx and feature sheets/components.
5. Instrument system/service metrics in src/services/* provider fetchers and event orchestration.
6. Add kill switch and sample-rate controls via Remote Config or env flags.

Core properties on every event

- app_version, build_number, platform, os_version
- tier (free/pro), auth_state (signed_in/signed_out), session_id
- city_scope fixed to London, feature_surface
- event_timestamp_utc generated client-side

3. Event taxonomy and naming

Naming convention

- User events: feature_action_result (example: route_preview_started, event_saved).
- System events: provider_status_metric (example: provider_timeout, events_cache_hit).
- Monetization events: paywall_action_step (example: paywall_shown, trial_started).
- No ambiguous names like click_1 or button_pressed.

Required identity model

- Before sign in: anonymous distinct_id from PostHog device identity.
- After sign in: identify(firebase_uid) and alias anonymous history.
- On sign out: reset identity to avoid cross-user contamination.
- Never use raw email as distinct_id.

High-value event families

- Onboarding: app_opened, splash_completed, product_tour_completed
- Discovery: filter_date_changed, filter_category_toggled, event_pin_tapped
- Navigation: route_preview_started, navigation_started, navigation_exited
- Retention: event_saved, flight_watch_saved, notification_channel_toggled
- Monetization: paywall_shown, paywall_dismissed, trial_started, subscription_started
- Reliability: provider_timeout, provider_failed, coverage_gap_detected

4. Feature instrumentation map (code-level)

Map and event discovery

Map exploration and event discovery

Priority events to instrument

- filter_date_changed (filter_key, result_count)
- filter_category_toggled (category, selected)
- event_pin_tapped (event_id, source, category)
- cluster_tapped (cluster_size)
- event_detail_opened and event_detail_closed
- layer_toggled (layer, enabled)

Primary hook files

- app/index.tsx
- src/components/FilterBar.tsx
- src/components/CategoryFilterBar.tsx
- src/components/EventDetailsSheet.tsx
- src/components/LayerControlPanel.tsx

Save and reminder flows

Priority events to instrument

- event_saved / event_unsaved
- event_reminder_scheduled
- calendar_add_started / calendar_add_success / calendar_add_failed
- saved_events_loaded (count)

Primary hook files

- app/index.tsx
- src/services/savedEvents.ts
- src/services/notifications.ts
- src/services/calendar.ts

Routing and navigation

Priority events to instrument

- route_preview_started (dest_kind)
- route_fetch_success / route_fetch_failed
- route_alternative_selected (index)
- navigation_started / navigation_exited
- navigation_off_route
- nav_app_selected (driveiq/google/waze/apple)

Primary hook files

- app/index.tsx
- src/services/routing.ts
- src/components/NavigationAppPicker.tsx
- src/components/RouteInfoPanel.tsx

Connections and station hubs

Priority events to instrument

- connections_panel_opened
- station_hub_attempt (station_id, result)
- free_station_claimed (station_id)
- line_detail_opened (line_id, severity)
- station_hub_navigate
- paywall_shown (feature=station_hubs)

Primary hook files

- app/index.tsx
- src/components/ConnectionsPanel.tsx
- src/components/StationHubSheet.tsx
- src/services/stationAccess.ts

Airports and flights

Priority events to instrument

- airport_pin_tapped (airport_id)
- airport_flights_sheet_opened
- flights_direction_changed
- flights_full_day_toggled
- flight_watch_saved / flight_watch_unsaved
- flight_watch_limit_hit (tier, limit)

Primary hook files

- app/index.tsx
- src/components/AirportFlightsSheet.tsx
- src/services/savedFlights.ts
- src/services/aerodatabox.ts

AI support and quotas

Priority events to instrument

- ai_support_opened
- ai_question_asked (answer_type only)
- ai_quota_consumed
- ai_quota_limit_hit
- ai_suggestion_tapped
- ai_action_remind and ai_action_calendar

Primary hook files

- src/components/AISupportSheet.tsx
- src/services/aiQuota.ts
- src/services/subscription.ts

5. Auth, paywall, and notifications

Auth instrumentation

- auth_sheet_opened (mode)
- auth_login_attempt/success/failed (error_code only)
- auth_signup_attempt/success/failed
- auth_reset_requested
- auth_logout
- account_profile_updated, account_email_updated, account_password_updated

Hook files: src/components/AuthSheet.tsx, src/providers/AuthProvider.tsx, src/components/AccountSheet.tsx

Paywall and entitlements

- Track paywall_shown centrally inside showProPaywall(feature).
- Track paywall_dismissed and paywall_cta_tapped from sheet/alert buttons.
- When RevenueCat is live, add trial_started, subscription_started, subscription_cancelled.
- Keep entitlement source of truth server-side; app only mirrors current tier for UX.

Notifications and alert quality

- notification_permission_requested/granted/denied
- notification_channel_toggled (channel, enabled)
- alert_road_incident_sent, alert_line_closure_sent, alert_flight_change_sent
- event_reminder_scheduled and event_end_reminder_scheduled

Hook files: src/services/notifications.ts, src/components/NotificationSettingsPanel.tsx, src/components/NotificationOnboarding.tsx

6. System reliability telemetry

Convert existing service logs into PostHog system events

- events_fetch_started / events_fetch_complete (total_count, sports_count)
- provider_success (provider, count, duration_ms)
- provider_failed (provider, error_type)
- provider_timeout (provider, timeout_ms)
- events_cache_hit/miss/stale, events_cache_written
- coverage_gap_detected (venue_labels)

Primary file: src/services/events.ts plus provider services in src/services/*

7. Privacy, PII, and compliance guardrails

Never send these to PostHog

- email, password, display name
- raw GPS coordinates, exact addresses
- chat question text or AI response text
- free-text feedback/report notes
- notification message body/title

Allowed identifiers

- firebase uid as identify key
- opaque event_id, flight_id, line_id, station_id
- booleans, buckets, counts, durations

Data minimization standard
If a property is not required for a dashboard decision, remove it. Prefer buckets over raw values and IDs over human text.

8. PostHog dashboard pack

Dashboard A - Activation

- app_opened -> event_pin_tapped conversion
- time_to_first_meaningful_action median
- first session save rate (event_saved or flight_watch_saved)

Dashboard B - Engagement

- WAU/DAU by tier
- feature usage mix: map, routing, flights, connections, AI
- navigation_started and event_saved trend by week

Dashboard C - Monetization

- paywall_shown -> paywall_cta_tapped
- trial_started -> subscription_started
- upgrade trigger attribution by feature

Dashboard D - Reliability

- provider_failed and provider_timeout by provider
- events total vs sports total trend
- coverage_gap_detected alerts over time

9. Implementation phases and QA gates

1. Phase 1: Core wrapper + identity + app lifecycle events
2. Phase 2: Map/routing/save flows + paywall hooks
3. Phase 3: Flights/connections/AI + quota and gate events
4. Phase 4: System telemetry + dashboards + data quality alerts

Pre-release QA checklist

- All events visible in PostHog live stream within 5 seconds in debug build
- No PII values appear in sampled payload inspection
- Anonymous-to-identified merge works after signup/login
- Signout resets identity and does not leak previous user timeline
- Every paywall surface emits paywall_shown with correct feature property
- Provider timeout simulation emits provider_timeout and does not crash app
- Dashboards produce stable numbers for 3 consecutive test runs

Operational tracking

- Assign one owner per feature area for instrumentation review.
- Keep event dictionary in repo and version with code changes.
- For every new feature PR, require analytics checklist completion.
- Review top 20 events monthly and prune low-value noise.

Immediate next step

Implement src/services/analytics.ts and wire the first 20 high-value events (auth, map taps, saves, routing start, paywall shown, provider timeout). Then validate dashboards before expanding scope.